Shijia Xu

M.S. Candidate in Control Science and Engineering

Chongqing University

shijiaxu@stu.cqu.edu.cn

+86 152 6558 4418

Personal Website

Google Scholar

Education

- **Queen Mary University of London** — *London, United Kingdom*
  Visiting Research Student in Computer Science — *Jun. 2026 – Dec. 2026*
  Host Supervisor: Prof. Ahmed M. A. Sayed *(Machine Learning Systems and Efficient Inference)*
  Funding: Supported by the Chongqing University Joint Training Program for Master's Students
  Research Focus: NLP and RAG for reliable and reasoning-capable Large Language Models
- **Chongqing University** — *Chongqing, China*
  M.E. Candidate in Control Science and Engineering*; GPA: 4.09/5.00 (**Rank: 2/74**)* — *Sep. 2024 – Jul. 2027*
  Advisor: Prof. Zhou Wu *(Optimization, Artificial Intelligence and Energy Systems)*
  Research Interests: Trustworthy Large Language Models, Retrieval-Augmented Generation, and Explainable NLP
  Selected Coursework: Graph Theory and Applications, Machine Learning and Big Data Processing, and Data Mining
- **Shandong University of Science and Technology** — *Qingdao, China*
  B.E. in Automation*; GPA: 4.03/5.00 (**Rank: 3/163**)* — *Sep. 2020 – Jul. 2024*
  Selected Coursework: Principles of Automatic Control, Intelligent Control, and Advanced Mathematics
  Selected Projects: ESN-Based Wind Energy Forecasting; FPGA-Based Wind Speed Measurement

Publication

- **Self-Correcting RAG: Enhancing Faithfulness via MMKP Context Selection and NLI-Guided MCTS**
  Shijia Xu, *Zhou Wu, Xiaolong Jia, Yu Wang, Kai Liu, April Xiaowen Dong.* — ***ACL 2026***
  - Introduces a unified RAG framework that formulates context selection as a Multi-dimensional Multiple-choice Knapsack Problem under a strict token budget and employs NLI-guided Monte Carlo Tree Search to explore and verify reasoning trajectories, improving reasoning accuracy while reducing hallucinations.
- **RCBSF: A Multi-Agent Framework for Automated Contract Revision via Stackelberg Game**
  Shijia Xu, *Yu Wang, Xiaolong Jia, Zhou Wu, Kai Liu, April Xiaowen Dong.* — ***ACL 2026***
  - Introduces a risk-constrained bilevel Stackelberg framework for automated contract revision, in which a global prescriptive agent imposes explicit risk budgets on revision and verification agents, enabling risk-aware optimization with theoretical convergence guarantees.
- **LLM-Guided Secure Federated Visual Prompts with Deep Unfolding for MRI Reconstruction**
  Di Xiao, Yuhan Gou, Yu Ren, ***Shijia Xu****, Yue Zhang.* — ***ICMR 2026***
  - Introduces FedLUR, a secure federated MRI reconstruction framework that combines deep unfolding, lightweight visual prompts, and privacy-preserving LLM-guided aggregation.
- **State Copying Crowds Out Reasoning: Mechanistic Evidence for Delta Planning in Autoregressive Models**
  Shijia Xu, *Wang Xi.* — ***NeurIPS 2026 (In Submission)***
  - Identifies copy–reason interference in autoregressive planning, showing that full-state generation disrupts goal-directed computation. Demonstrates through controlled experiments and activation patching that delta-style planning substantially improves long-horizon reasoning.

- **Resource-Aware Federated LoRA Fine-Tuning for Heterogeneous Environments**
  *Xiaolong Jia, Ahmed M. Abdelmoniem, **Shijia Xu**, Gaoyang Liu, Chen Wang.* ***NeurIPS 2026 (In Submission)***
  - Proposes resource-aware federated fine-tuning with budget-adaptive LoRA ranks, Top-k sparsification, and selective regularization for efficient communication and robust adaptation.
- **The Cost of Compression: A Rate-Distortion Limit on Factual Hallucination**
  Wang Xi, **Shijia Xu** (co-first author), Rongfeng Guo.* ***EMNLP 2026 (In Submission)***
  - Establishes an information-theoretic coverage–compression framework that decomposes factual hallucination into compression distortion on observed facts and missing coverage on unseen facts.
- **MemSIF: From Structured Interactions to Dual-Track Fact Memory for LLM Agents**
  *Yufei Luo, **Shijia Xu**, Guangyuan Dong, Xiucheng Xu.* ***AAAI 2027 (In Submission)***
  - Introduces MemSIF, combining structured interaction memory with CoreFact and ActiveFact tracks to improve long-term agent memory on LoCoMo and LongMemEval-S.

Project Experience

- **Intelligent Contract Management System**
  Core Developer and Project Lead *Jan. 2025 – Oct. 2025*
  - Developed a full-stack contract management platform using Flask and Redis, featuring role-based access control, real-time communication, document versioning, and template management. Integrated DeepSeek LLM for automated contract generation, legal risk analysis, and clause extraction.
- **Robot Team, Shandong University of Science and Technology**
  Team Leader and Head of the Control & Electronics Group *Dec. 2021 – May 2023*
  - Led a 30-member team in the ABU Robocon competition, coordinating robot development and competition strategy. Oversaw embedded control programming, hardware design, sensor integration, and system-level debugging for manual and autonomous robots.

Honors

- **[1] Chongqing University First Prize Master's Academic Scholarship** *(2024 & 2025)*
- **[2] Chongqing University Merit Student** *(2025)*
- **[3] Outstanding Graduate of Shandong Province** *(2024)*
- **[4] Shandong Provincial Government Scholarship** *(2022)*
- **[5] Sun Yueqi Outstanding Student Award** *(Top Industry Honor)* *(2022)*
- **[6] National Endeavor Scholarship** *(Top 3%)* *(2021)*

Awards

- **[1]** 15^{th} **MathorCup Math Application Challenge (Postgraduate) - National Second Prize** *(2025)*
- **[2] RAICOM Developer Robot Competition - National Second Prize** *(2023)*
- **[3] Nat'l Undergrad Robotics Competition (ROBOCON) - National Third Prize** *(2022 & 2023)*
- **[4] Int'l Advanced Robot & Simulation Competition - National Second Prize** *(2022)*
- **[5] MathorCup Math Modeling Challenge - National Third Prize** *(2022)*
- **[6] Intelligent Control Contest - Provincial First Prize** *(2022)*
- **[7] Nat'l Undergrad Electronics Design Contests - Provincial Second Prize** *(2021 & 2022)*

Professional Skills

- **Programming Skills:** Proficient in Python, Java, SQL, Excel, PowerPoint; Experienced in PyTorch and TensorFlow machine learning development; Intermediate in MATLAB, C; Relevant projects in NLP and CV.
- **Interests:** Texas Hold'em Poker, travel blogging, photography, and basketball.